ELECTRONIC PRODUCT DESIGN & REVERSE ENGINEERING

Project Statements | Part B | Q. No. 3b

Full-Wave Rectifier

PN Junction Diode Bridge: Simulation & Comparison with Half-Wave Rectifier



SUBMISSION DETAILS

Group Number: 53

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1 Problem Statement

Q. No.	Problem Description
3b	Simulate a full-wave rectifier using a PN junction diode. Infer its behavior and explain its rectification efficiency and ripple factor in comparison to a half-wave rectifier.

Objective

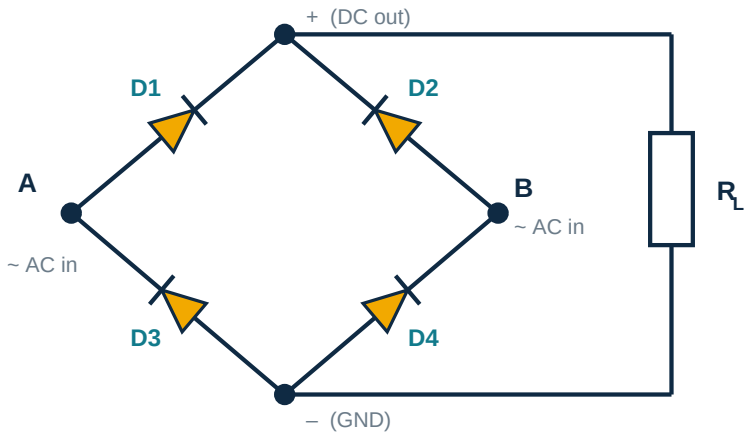
- Build and simulate a full-wave bridge rectifier using four PN junction diodes.
- Observe the rectified output on an oscilloscope and infer how the circuit behaves.
- Compare rectification efficiency (η) and ripple factor (γ) with a half-wave rectifier.

Components and Instruments

Item	Qty	Purpose
PN junction diodes (silicon)	4	Form the bridge
Load resistor R_L	1	Load across the DC output
Function generator	1	AC source (frequency, amplitude, offset)
Oscilloscope	1	Displays the output waveform across R_L
Breadboard and jumper wires	–	Circuit assembly in the simulator

2 Theory: Bridge Rectifier Operation

A full-wave bridge rectifier uses four diodes so that **both half-cycles** of the AC input drive current through the load in the **same direction**. In each half-cycle exactly two diodes are forward biased and the other two are reverse biased.



Positive half-cycle (A positive)
D1 and D4 conduct: A → D1 → R_L → D4 → B.

Negative half-cycle (B positive)
D2 and D3 conduct: B → D2 → R_L → D3 → A.

In both cases current flows through R_L from the + node to the – node, so the output never reverses polarity.

Figure 1: Bridge rectifier circuit (diodes D1–D4, load R_L).

Output Waveforms

The figure below contrasts the ideal outputs. The half-wave rectifier discards every negative half-cycle, while the bridge folds it over, so the output pulses occur at **twice the input frequency**.

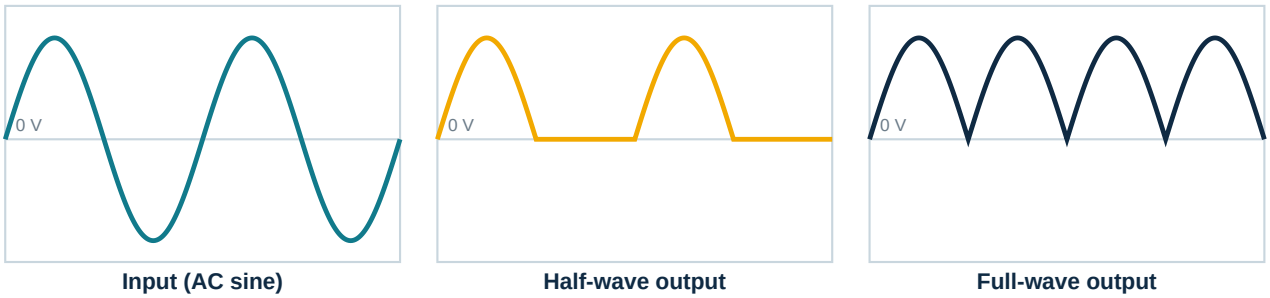


Figure 2: Ideal input, half-wave output and full-wave output (two input cycles).

3 Circuit Simulation

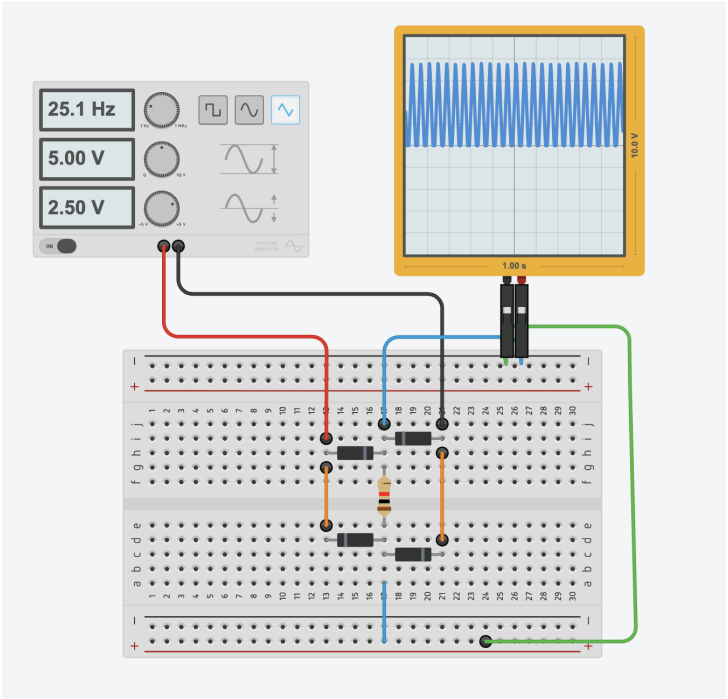


Figure 3: Simulated bridge rectifier on a breadboard, with function generator and oscilloscope.

Setup Description

- The four diodes are placed on the breadboard as a bridge. The function generator is connected across the two AC corners of the bridge (columns 13 and 21).
- The load resistor sits between the bridge's DC corners (column 17, upper and lower halves).
- The oscilloscope probes are connected across the load resistor, so the trace shows the rectified output voltage.

Source and Oscilloscope Settings (from Figure 3)

Parameter	Setting
Frequency	25.1 Hz
Amplitude	5.00 V
DC offset	2.50 V
Oscilloscope time base	1.00 s (full screen width)

4 Observations and Inference

- The oscilloscope trace stays on one side of the zero line and consists of a dense train of identical positive pulses. There are **no negative half-cycles**, which confirms full-wave rectification.
- For a 25.1 Hz source the output ripple frequency is **2 × 25.1 = 50.2 Hz**, double that of a half-wave rectifier on the same input. In a 1.00 s window this means about 50 pulses.
- Two diodes conduct at any instant, so the output peak is lower than the input peak by about **2 × 0.7 V = 1.4 V** for silicon diodes.
- The output is unidirectional but still pulsating. Without a filter capacitor the ripple remains large, and the DC level is the average of the pulses, not the peak.

> **Note on the source settings:** the formulas in this report assume a **sinusoidal input with 0 V DC offset**. Figure 3 shows an offset of 2.50 V, which shifts the input and makes alternate output pulses unequal. For an ideal symmetric result, set the offset to 0 V and select the sine waveform before taking measurements.

5 Efficiency and Ripple Factor Analysis

Rectification Efficiency (η)

Efficiency is the ratio of DC output power delivered to the load to the AC input power. For an ideal diode and a sinusoidal input:

$$\eta = P_{dc} / P_{ac} = V_{dc}^2 / V_{rms}^2$$

- **Half-wave:** $V_{dc} = V_m/\pi$ and $V_{rms} = V_m/2$, so $\eta = 4/\pi^2 \approx$ **40.6 %**.
- **Full-wave:** $V_{dc} = 2V_m/\pi$ and $V_{rms} = V_m/\sqrt{2}$, so $\eta = 8/\pi^2 \approx$ **81.2 %**.

Ripple Factor (γ)

The ripple factor measures how much AC remains in the output relative to its DC value:

$$\gamma = V_{ac,rms} / V_{dc} = \sqrt{(F^2 - 1)}, \text{ where } F = V_{rms} / V_{dc} \text{ is the form factor}$$

- **Half-wave:** $F = \pi/2 = 1.57$, so $\gamma \approx$ **1.21**.
- **Full-wave:** $F = \pi/(2\sqrt{2}) = 1.11$, so $\gamma \approx$ **0.482**.

The full-wave output is therefore about 2.5 times less rippled, and its ripple frequency (2f) is easier to smooth with a smaller filter capacitor.

6 Half-Wave vs Full-Wave Comparison

Parameter	Half-wave	Full-wave (bridge)
Number of diodes	1	4
DC output voltage V_{dc}	$V_m/\pi \approx 0.318\ V_m$	$2V_m/\pi \approx 0.636\ V_m$
RMS output voltage V_{rms}	$V_m/2 = 0.5\ V_m$	$V_m/\sqrt{2} \approx 0.707\ V_m$
Form factor F	1.57	1.11
Ripple factor γ	1.21	0.482
Rectification efficiency η	40.6 %	81.2 %
Ripple frequency	f (25.1 Hz)	2f (50.2 Hz)
Peak inverse voltage per diode	V_m	V_m
Diode drops in the current path	1 (≈ 0.7 V)	2 (≈ 1.4 V)

Expected Output for This Setup

Taking V_m = 5 V, silicon diodes (0.7 V drop), a sinusoidal input and zero offset:

Quantity	Half-wave	Full-wave (bridge)
Peak output voltage	5 – 0.7 = 4.3 V	5 – 1.4 = 3.6 V
Average (DC) output	4.3/π ≈ 1.37 V	2 × 3.6/π ≈ 2.29 V

The bridge gives about **1.7 times** the DC output of the half-wave circuit, even after paying the extra diode drop. Because V_m is small, the diode drops lower the practical efficiency below the ideal 81.2 % and 40.6 %.

7 Conclusion

- The bridge rectifier converts both halves of the AC input into unidirectional pulses, as seen on the oscilloscope.
- Its ideal rectification efficiency is **81.2 %**, double the 40.6 % of a half-wave rectifier.
- Its ripple factor is **0.482** against 1.21, and the ripple frequency doubles to 2f, which makes filtering easier.
- The cost is four diodes instead of one and an extra ≈ 0.7 V drop, which matters at low supply voltages.
- Adding a filter capacitor across R_L would reduce the ripple further.